

Test Plan and Test Cases: Alumni Mentoring Portal - Registration Module

1. Master Test Plan (Brief)

- **Objective:** To verify the functionality and reliability of the Registration module for the Alumni Mentoring Portal, ensuring data integrity and adherence to manual validation constraints.
- **Scope:** This plan covers the Student and Alumni registration processes, including UI validation and backend database storage in MySQL.
- **Testing Approach:** Manual testing focusing on input validation (HTML/CSS/JS) and successful database entry via Servlets/CGI/Socket programming. No external libraries are used for validation or duplicate checking.
- **Roles:** Quality Assurance (QA) team (Team D per Week 1-3 rotation).

2. Severity Classification

Level	Description
High	Major functional failure where valid registration data is rejected by the system.
Medium	Minor functional issues or UI bugs that do not prevent the core registration flow.
Low	Minor cosmetic, formatting, or UI alignment issues on registration pages.

3. Test Cases for Registration Module

The following test cases follow a manual execution strategy as required by the implementation guidelines.

TC ID	Description & Pre-conditions	Input Data	Steps & Expected Result	Severity
REG-01	Positive: Valid Student Registration. <i>Pre-condition:</i> DB connection active.	Name: Parth Chaudhari, Email: parthrc7@gmail.com, Pass: Study@123	1. Enter details. 2. Click Register. <i>Result:</i> Data stored in MySQL; Success message shown.	High
REG-02	Positive: Valid Alumni Registration. <i>Pre-condition:</i> Registration page loaded.	Name: John Doe, Email: john.alumni@test.com, Pass: Alumni#2024	1. Fill Alumni form. 2. Submit. <i>Result:</i> Alumni record created in DB successfully.	High
REG-03	Negative: Empty Mandatory Fields. <i>Pre-condition:</i> Fields are empty.	Name: "", Email: "", Password: ""	1. Click Register. <i>Result:</i> UI displays error "Mandatory fields cannot be empty".	Medium
REG-04	Negative: Invalid Email Format. <i>Pre-condition:</i> Registration page active.	Email: parthrc7.gmail.com (missing @)	1. Enter invalid email. 2. Click Register. <i>Result:</i> JS/Backend validation rejects input.	Medium
REG-05	Negative: Password Mismatch. <i>Pre-condition:</i> Form filled.	Pass: Admin123, Confirm Pass: Admin456	1. Enter non-matching passwords. 2. Click Register. <i>Result:</i> Validation error shown.	Medium
REG-06	Negative: Duplicate Registration. <i>Pre-condition:</i> Email already exists in DB.	Email: parthrc7@gmail.com	1. Enter existing email. 2. Click Register. <i>Result:</i> Manual duplicate check identifies existing user.	High
REG-07	Negative: Password Length Constraint.	Password: "pass" (4 chars)	1. Enter short password. 2. Click Register. <i>Result:</i>	Medium

TC ID	Description & Pre-conditions	Input Data	Steps & Expected Result	Severity
	<i>Pre-condition:</i> Min length is 8 chars.		Validation error for minimum character limit.	

TC ID	Description & Pre-conditions	Input Data	Steps & Expected Result	Severity
REG-08	Negative: Invalid Email Format (No domain). <i>Pre-condition:</i> Registration page active.	Email: test@	1. Enter email. 2. Click Register. <i>Result:</i> Validation error for missing domain.	Medium ▾
REG-09	Negative: Special Characters in Name. <i>Pre-condition:</i> Registration page active.	Name: Parth@123	1. Enter name. 2. Click Register. <i>Result:</i> Validation error for invalid characters.	Low ▾
REG-10	Negative: SQL Injection Attempt in Password. <i>Pre-condition:</i> Registration page active.	Pass: ' OR 1=1 --	1. Enter payload in password field. 2. Click Register. <i>Result:</i> System rejects input/sanitizes properly.	High ▾
REG-11	Negative: Field Length Exceeded (Name). <i>Pre-condition:</i> Registration page active.	Name: [101 character string]	1. Enter long name. 2. Click Register. <i>Result:</i> Error message regarding character limit.	Low ▾
REG-12	Negative: Whitespace Only Input. <i>Pre-condition:</i> Registration page active.	Name: " "	1. Enter spaces. 2. Click Register. <i>Result:</i> Validation error for mandatory fields.	Medium ▾

TC ID	Description & Pre-conditions	Input Data	Steps & Expected Result	Severity
REG-13	Negative: Invalid Phone Number format. <i>Pre-condition:</i> Registration page active.	Input: 'abc1234567'	1. Enter phone number. 2. Click Register. <i>Result:</i> Error message for invalid numeric format.	Medium ▾
REG-14	Negative: Underage Registration. <i>Pre-condition:</i> Age constraint rule active.	Input: DOB < 18 years	1. Select DOB. 2. Click Register. <i>Result:</i> Validation error for age eligibility.	Low ▾
REG-15	Negative: Username/Name Length Too Short. <i>Pre-condition:</i> Min 3 chars.	Input: 'AB'	1. Enter name. 2. Click Register. <i>Result:</i> Error for minimum length.	Low ▾
REG-16	Negative: HTML Tag Injection. <i>Pre-condition:</i> Registration page active.	Input: '<script>alert(1)</script>'	1. Enter payload. 2. Click Register. <i>Result:</i> Input sanitized or blocked.	High ▾
REG-17	Negative: Missing Confirm Password. <i>Pre-condition:</i> Form partially filled.	Input: Pass='Secret123', Confirm=''	1. Enter details. 2. Click Register. <i>Result:</i> Validation error for matching fields.	Medium ▾

TC ID	Description & Pre-conditions	Input Data	Steps & Expected Result	Severity
REG-18	Negative: Invalid Date of Birth Format. <i>Pre-condition:</i> Registration page active.	Input: '31-02-2024' (Invalid date)	1. Enter invalid date. 2. Click Register. <i>Result:</i> Error message regarding invalid date format.	Low ▾

TC ID	Description & Pre-conditions	Input Data	Steps & Expected Result	Severity
REG-19	Negative: Password containing only digits. <i>Pre-condition:</i> Registration page active.	Input: '12345678'	1. Enter numeric password. 2. Click Register. <i>Result:</i> Error message requiring alphanumeric password.	Medium ▾
REG-20	Negative: Email with consecutive dots. <i>Pre-condition:</i> Registration page active.	Input: 'test..user@domain.com'	1. Enter invalid email. 2. Click Register. <i>Result:</i> Validation error for invalid email format.	Medium ▾
REG-21	Negative: Empty Name Field. <i>Pre-condition:</i> Registration page active.	Input: Name = ""	1. Leave name blank. 2. Click Register. <i>Result:</i> "Name is mandatory" validation error.	Medium ▾
REG-22	Negative: Unsupported special characters in Address. <i>Pre-condition:</i> Registration page active.	Input: 'Address with \'\'	1. Enter characters. 2. Click Register. <i>Result:</i> Validation error for invalid characters in address.	Low ▾
REG-23	Negative: Exceeding Max length of Password field. <i>Pre-condition:</i> Registration page active.	Input: 50+ chars string	1. Enter long password. 2. Click Register. <i>Result:</i> Truncation or error message regarding max length.	Medium ▾
REG-24	Negative: Registration with future Date of Birth. <i>Pre-condition:</i> Registration page active.	Input: '20-10-2030'	1. Enter future DOB. 2. Click Register. <i>Result:</i> Validation error stating DOB cannot be in future.	Medium ▾

TC ID	Description & Pre-conditions	Input Data	Steps & Expected Result	Severity
REG-25	Negative: Username with leading spaces. <i>Pre-condition:</i> Registration page active.	Input: ' Parth'	1. Enter name with spaces. 2. Click Register. <i>Result:</i> Error message or input sanitization (trimming).	Medium ▾
REG-26	Negative: Unchecked Terms and Conditions. <i>Pre-condition:</i> Registration page active.	Input: Checkbox = Unchecked	1. Fill details but leave T&C unchecked. 2. Click Register. <i>Result:</i> Error message to accept terms.	Medium ▾
REG-27	Negative: SQL Injection in Email field. <i>Pre-condition:</i> Registration page active.	Input: 'admin@site.com' OR 1=1--	1. Enter SQL payload. 2. Click Register. <i>Result:</i> System rejects input and sanitizes fields.	High ▾

4. Conclusion

This test suite provides a comprehensive validation framework for the Alumni Mentoring Portal's Registration module. By covering a broad spectrum of scenarios - including positive functional flows, negative boundary condition testing, data integrity checks & critical security validation (such as SQL and XSS injection prevention) - this document ensures that the registration process is robust, secure & user-friendly. These manual test cases serve as a foundational quality assurance guide to maintain data consistency and prevent system failures during user onboarding.